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ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Week 2: Mechanical Design & Design Requirements

Overview / Outline



Today's Topics

- Translating a problem into **design requirements**
- Designing for **assembly & manufacturing**
- Selecting and sizing **gears**
- Common **mechanisms & power transmission**
- Designing with **linkages**
- **Joints, bearings, and fasteners**
- Basic **center of mass & load considerations**

Goal: Learn how to choose mechanical components and mechanisms based on what your robot actually needs to accomplish

Design Requirements (Scoping)



Requirement Type	Question to Ask	Example
Functional	What must it do?	Lift a 5 kg object
Performance	How accurate/fast?	Lift 5 kg in <3 sec
Constraint	What limits the design?	<4 kg, <\$100, fits in 12" × 12"
Interface	What must it connect to?	Mount to existing chassis

Good requirements should be measurable.

Instead of:

"Robot arm should be strong."

Use:

"Robot arm shall lift a 2 kg payload at 300 mm extension."

Design Requirements: Don't Overdesign



“Simple + reliable” usually beats “complex + unfinished”.

- Design for the requirements, not for every possible scenario.
- Reduce unnecessary parts and degrees of freedom.
- Use proven mechanisms when possible.
- Identify likely failure points before building.
- Leave room for iteration.

Ask Before You CAD

1. What does it need to do?
2. What loads will it experience?
3. What is the simplest mechanism that accomplishes the task?
4. How will it be manufactured?
5. How will it be assembled and repaired?
6. What is most likely to fail?

Design for Assembly & Manufacturing



Design for Assembly

- Minimize part count
- Make assembly intuitive
- Minimize the number of tools required
- Make frequently serviced components accessible
- Use modular/subassembly-based designs
- Avoid inaccessible screws and fasteners

Design for Manufacturing

- Can we actually manufacture it?
- What machine/process will make it?
- Does the geometry work with that process?
- How long will manufacturing take?
- What tolerances are actually necessary?
- Can a simpler geometry accomplish the same thing?

CAD-able ≠ Manufacturable

Generalized COM and Momentum Analysis



Center of Mass (Full Assembly): $\bar{x} = \frac{m_1x_1 + m_2x_2 + \dots + m_nx_n}{m_{total}}$

Rotational Kinetic Energy: $KE = \frac{1}{2}I\omega^2$ $I = \sum m_i r_i^2$

- Higher I (mass at rim) and higher ω (RPM) = more destructive energy

Angular Momentum: $L = I\omega$

- Causes robot to pitch or roll when turning (worse at high RPM)
- Spinning weapons resist changes in orientation

Why it Matters:

- Front-heavy load prone to tipping (COM outside support base)
- Unbalanced COM will have an uneven load on drive system
- Hit reaction: at COM (translation) vs. away from COM (translation & rotation)

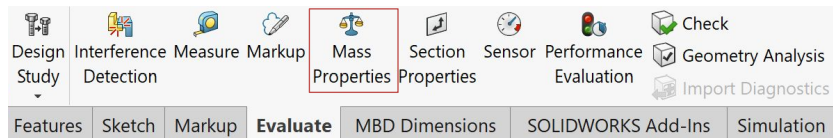
Rule of Thumb: Keep COM low and centered for predictable/reliable movement

CAD COM Assembly Evaluation



Accessing Mass Properties:

1. Open Assembly (.SLDASM) file
2. Tools → **Mass Properties** (or Evaluate tab → Mass Properties)
3. Solidworks calculates COM for the entire assembly automatically

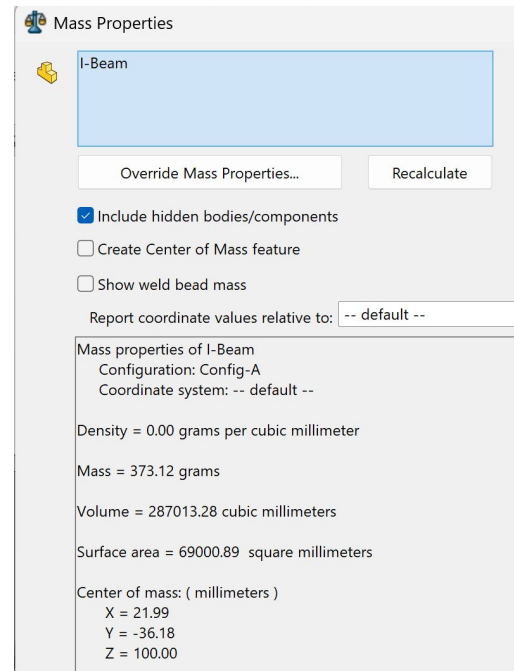


Setup for Accurate Results:

- Assign correct materials to every part (Right-click part → Material)
- Set your coordinate origin intentionally

Overriding Mass Properties (Per Part)

1. Tools → Mass Properties → Override Mass Properties
2. Manually enter mass (weigh the actual printed part [g])
3. COM location (measure or estimate from geometry)





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Gears

Types of Gears (Parallel Shafts)



Spur Gear

Teeth: Straight, parallel to shaft

Pros: Simple, efficient, inexpensive

Cons: Noisy at high speeds

Robotics: Gearboxes, drivetrains, mechanisms

Helical Gear

Teeth: Angled relative to shaft

Pros: Smooth, quiet, greater tooth engagement

Cons: Generates axial thrust; harder to manufacture

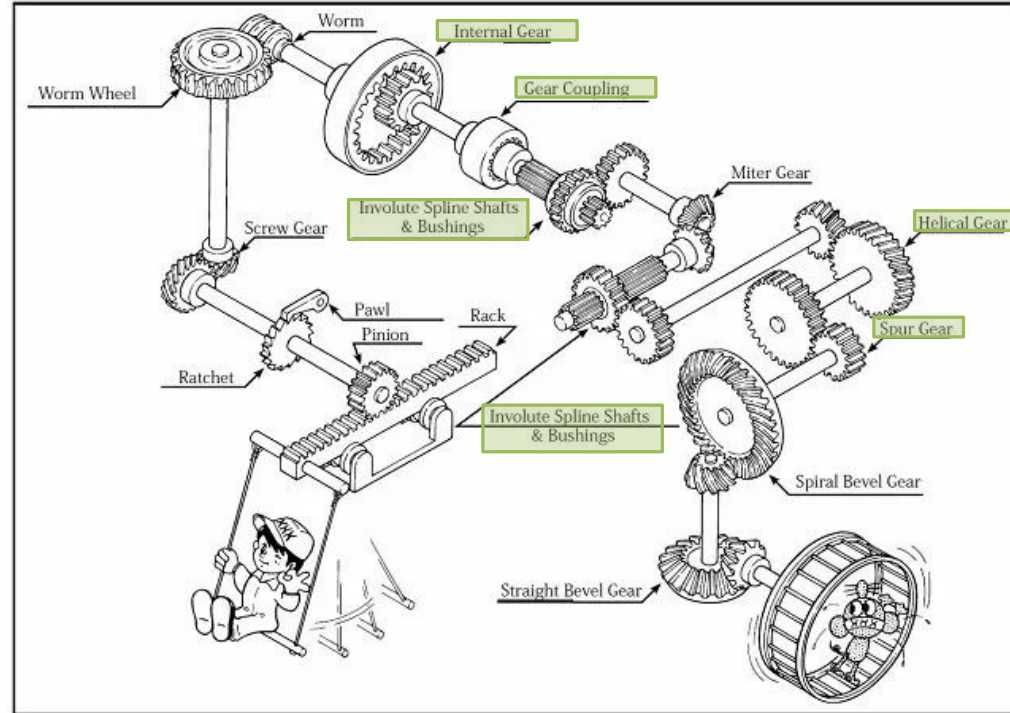
Robotics: Higher-load/smoother gearboxes

Internal Gear

Teeth: Located on inside of a ring

Pros: Compact, useful for high reduction systems

Robotics: Planetary gearboxes



Types of Gears (Non-Parallel Shafts)



Bevel Gear

Transfers rotation between **intersecting shafts**, commonly 90° .

Application: right-angle drivetrains.

Spiral Bevel Gear

Bevel gear with curved/angled teeth.

Pros: smoother, quieter, higher load capability

Cons: more difficult to manufacture

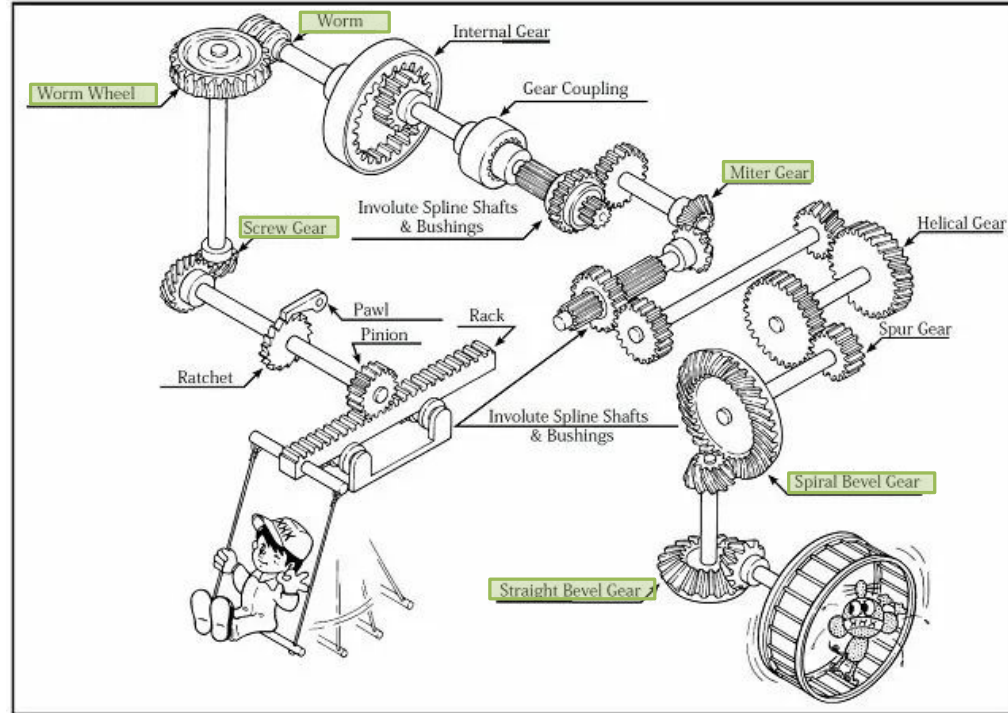
Worm Gear

A worm drives a worm wheel, usually at 90° .

Pros: Very large reduction in one stage, compact, resist back-driving

Cons: High sliding friction, lower efficiency, heat

Applications: lifts, positioning mechanisms, high-reduction systems.



Planetary Gearbox



Components:

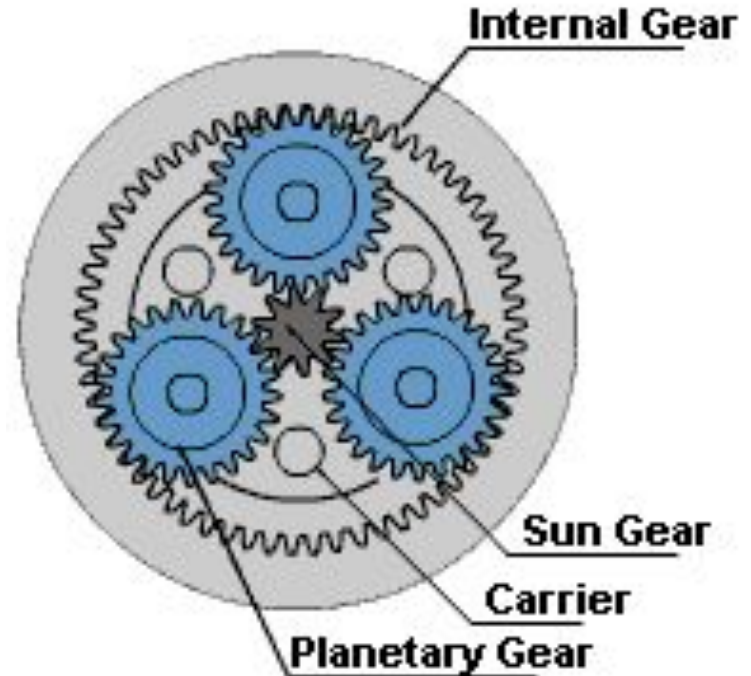
- **Sun gear** — center
- **Planet gears** — rotate around sun
- **Planet carrier** — holds planets
- **Ring gear** — internal outer gear

Advantages

- Large reduction in a **compact package**
- High torque density
- Input/output can remain coaxial
- Load distributed across multiple planet gears

Common Robotics Application:

Planetary gearbox attached to a DC or brushless motor.



Cycloidal Gearbox



Components:

- Eccentric input / bearing
- Cycloidal disc(s)
- Ring pins / rollers
- Output pins / rollers
- Output shaft or carrier

Advantages

- Near zero backlash
- Low Back-drivability & high holding resistance
- Strong shock load tolerance
- High torque density
- Input/output can remain coaxial

Common Robotics Application:

- High-torque robotic joints & actuators

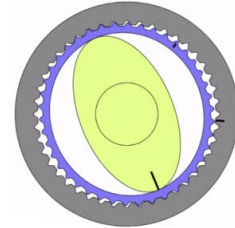


Harmonic Drive (Strainwave Gearbox)



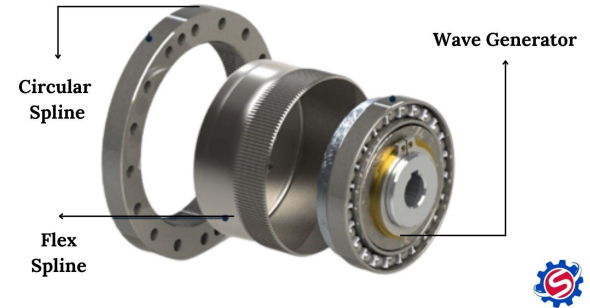
Components:

- **Wave Generator** — elliptical input that deforms the flexspline
- **Flexspline** — flexible gear with external teeth
- **Circular Spline** — rigid outer gear with internal teeth



Advantages

- Low backlash & high positioning precision
- Compact and lightweight
- Highly **back-drivable**
- High torque density
- Input/output can remain coaxial



Common Robotics Application:

- Precision robot arm joints & collaborative robot actuators

Gearbox Comparison Table



Property	Planetary	Cycloidal	Harmonic Drive
Backlash	Low	Near Zero	Zero
Shock Load Tolerance	Moderate	Excellent	Poor
Back-drivability*	Moderate	Low	High
Torque density	High	High	Very High
Noise	Moderate	Low	Very Low
Stiffness	High	High	Lower (flex element)
Cost	Low	Medium	High

**high/low benefit depends on application*

Gear Couplings



Purpose: Connect two shafts so **torque and rotational motion** can be transmitted between them.

How They Work

- External gear teeth on shaft hubs mesh with internal teeth in a sleeve.
- Allows small amounts of **shaft misalignment** while still transmitting torque.

Advantages

- High torque capacity
- Compact
- Can tolerate slight angular/parallel misalignment

Applications

- Motors → gearboxes
- Drivetrain connections
- Industrial machinery
- High-torque rotating systems

A gear coupling *connects* shafts. It is not primarily used to change speed or torque like a gear train.

Involute Splines



Purpose: A tooth shape/profile used to connect a shaft to a rotating component while transmitting high torque.

How They Work

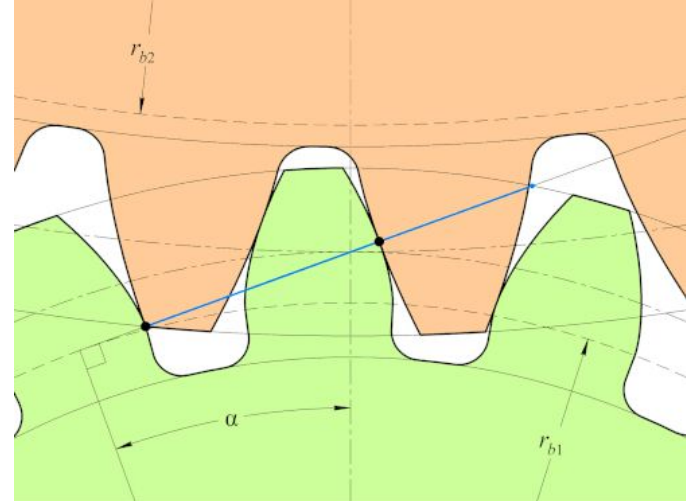
- Multiple teeth are distributed around the shaft.
- Load is distributed across multiple teeth instead of one key or flat.

Advantages

- High torque capacity
- Good alignment between components
- Stronger load distribution
- Some designs allow axial sliding

Applications

- Gearbox shafts
- Automotive drivetrains
- Robotics transmissions



Pressure Angle & Backlash



Pressure Angle: The angle at which force is transmitted between two meshing gear teeth.

Contact force can be separated into:

- **Tangential force** → produces useful torque
- **Radial force** → pushes the gears apart and loads the shafts/bearings

Larger Pressure Angle:

- Stronger gear teeth
- Greater load capacity
- Greater radial/separating force

Smaller Pressure Angle:

- Smoother, quieter operation
- Lower separating force
- Weaker tooth profile

Common: 20° is the standard general-purpose pressure angle.

Backlash: The amount the input gear can rotate before the output gear begins moving when reversing direction.

Why have backlash?

- Prevents gears from binding
- Allows for manufacturing tolerances
- Provides space for lubrication
- Accounts for thermal expansion

Too Much Backlash

- Reduced positioning accuracy
- Impact between teeth
- Poor precision control

Too Little Backlash

- Increased friction
- Increased wear
- Possible overheating

Gear Ratio and RPM Formula



Output RPM = Input RPM x Gear Ratio

$$R = \frac{Teeth_{Output}}{Teeth_{Input}} = \frac{Radius_{Driven}}{Radius_{Driver}}$$

$$R_{ab} * R_{bc} = R_{bc}$$

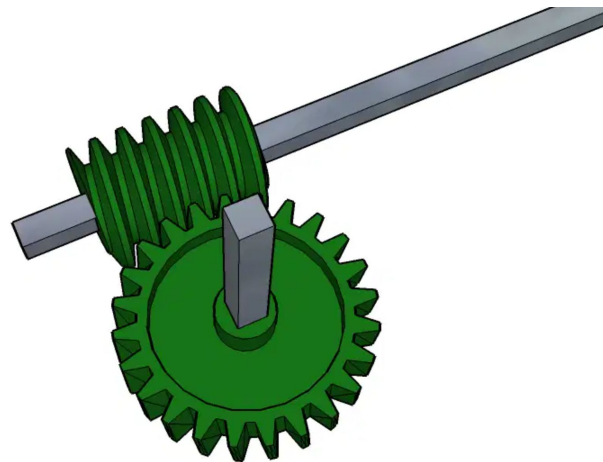
Gear Ratio Practice (1)



Examine the image of the gear combination and determine the end ratio. Express this ratio as a single number.

Gears = worm and 24T helical

- a. 3.4
- b. 168
- c. 72
- d. 24



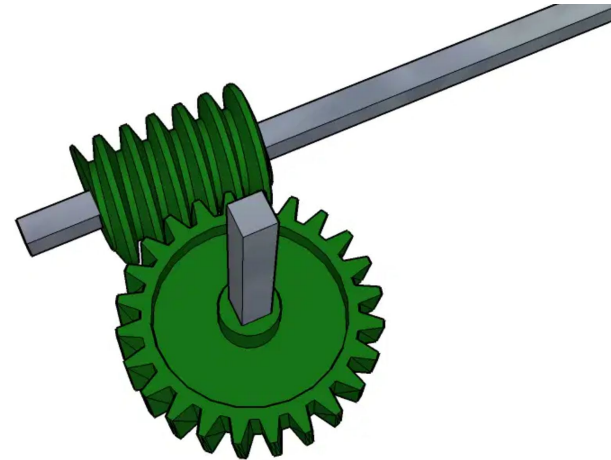
Gear Ratio Practice (1)



$$\text{Gear Ratio} = \frac{\text{Driven Gear Teeth}}{\text{Worm Starts}}$$

$$\text{Gear Ratio} = \frac{24}{1} = 24$$

The worm gear must rotate 24 times for the driven gear to rotate once.



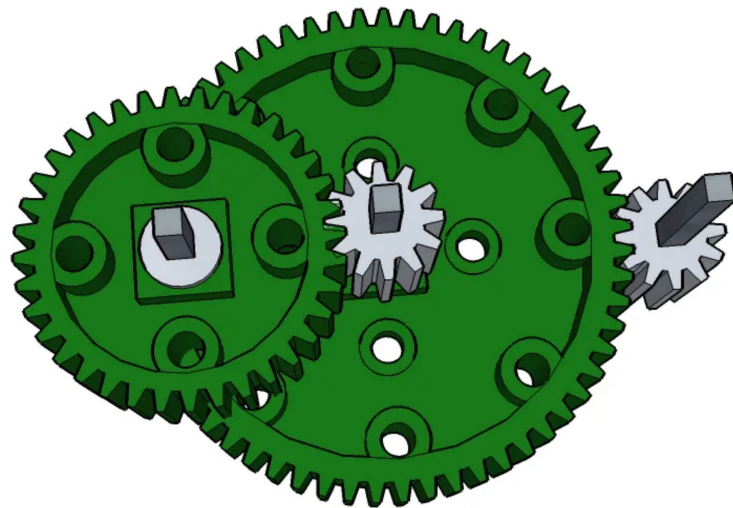
Gear Ratio Practice (2)



Examine the image of the gear combination and determine the end ratio. Express this ratio as a single number.

gear tooth # = 12, 36, 60

- a. 8
- b. 12
- c. 15
- d. 120



Gear Ratio Practice (2)

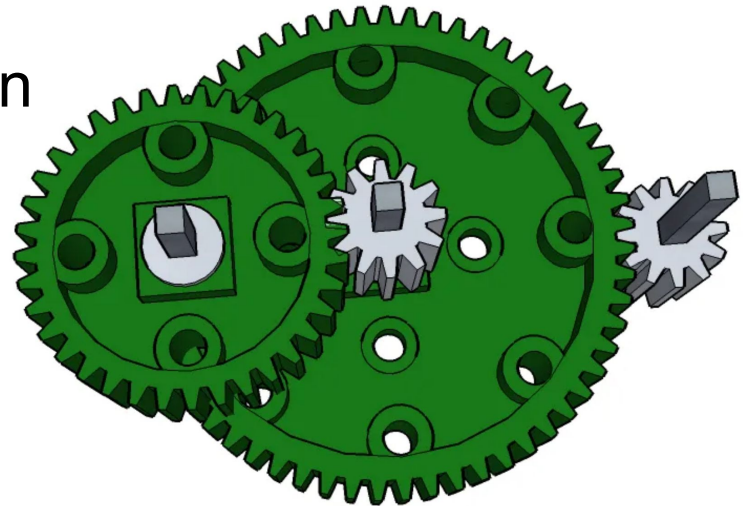


$$\text{Stage 1: Gear Ratio} = \frac{36T}{12T} = 3$$

$$\text{Stage 2: Gear Ratio} = \frac{60T}{12T} = 5$$

$$\text{Overall Gear Ratio} = 3 \times 5 = 15$$

If the driver gear spins 15 times, then the last driven gear spins once



Gear CAD Database & Generator



full gear CAD database: rushgears.com

Part Search and CAD

Partno:

[Click To Build CAD Models](#)

[Click To Retrieve Your Parts](#)

Pitch Type: Gear Type: Material: Units: Pitch: Teeth: PA: Style: ☒ auto get specs

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 > >>

Total Pages: 44

CAD	HT	PARTNO	CUSTOMIZE	QUOTE	PITCH	TEETH	PD	OD	PA	STYLE	BORE	FACE	HD	HP	OAL
CAD	HP	S1011	Customize	☐	10	11	1.2	1.4	14.5	B	0.625	1	0.94	0.63	1.63
CAD	HP	F1011	Customize	☐	10	11	1.2	1.4	20	A	0.625	1	--	--	1
CAD	HP	A1011	Customize	☐	10	11	1.2	1.4	14.5	A	0.625	1	--	--	1





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Common Mechanisms

Rack & Pinion



Components: Uses a circular pinion gear and linear gear rack to convert rotational motion ↔ linear motion.

Speed: Larger pinions produce more linear travel per revolution; smaller pinions produce less travel with greater mechanical advantage.

Precision: Provides direct, predictable motion, but backlash between the teeth can reduce positioning accuracy.

Support: Rack requires proper guides/supports to maintain alignment and tooth engagement.

Common Applications: Steering systems, linear actuators, gantries, sliding mechanisms, and robotic positioning systems.



Rack Travel per Revolution = $\pi \times \text{Pinion Pitch Diameter}$

Lead Screw & Ball Screw



Lead Screw

Function: Converts rotational motion into linear motion with a threaded screw & nut

Advantages: Simple, inexpensive, & can hold high loads

Drawbacks: More friction & lower efficiency than a ball screw.

Applications: 3d Printers, vises, linear actuators, & lifts

Ball Screw

Function: Converts rotational motion into linear motion using a screw, nut, and ball bearings.

Advantages: High efficiency, precision, & smooth motion.

Drawbacks: More \$\$\$

Applications: CNC machines, automation, & precision machining



Chain & Belt Drives



Chain Drive

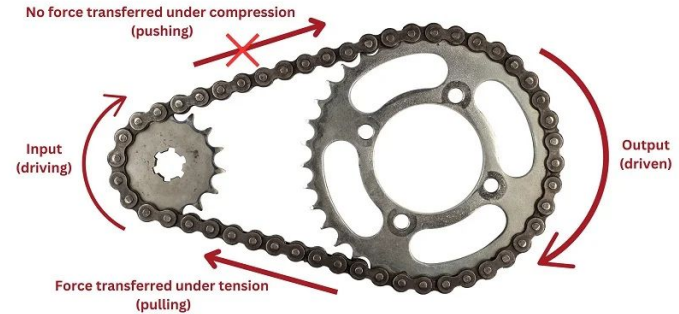
Components: Uses chains & sprockets to transfer rotational motion.

Engagement: Teeth prevent slipping between the chain & sprocket.

Strength: Utilized in high load and torque applications.

Maintenance: Requires occasional maintenance & lubrication.

Ratio: Driven sprocket teeth / Driver sprocket teeth.



Belt Drive

Components: Uses a belt & pulley to transfer rotational motion.

Operation: Provides quieter and smoother motion than a chain drive.

Distance: Can efficiently transfer motion between shafts far apart

Slippage: Possible slippage under heavy loads.

Ratio: Driven Pulley Diameter / Driver Pulley Diameter



Pulleys



Pulleys guide or transmit force/motion using a belt, rope, or cable around a rotating wheel.

Straight Pulley: Flat surface commonly used with belts that are mechanically guided or otherwise kept aligned.

Crowned Pulley: Pulley diameter is slightly larger at the center, helping a flat belt self-center instead of walking off the pulley.

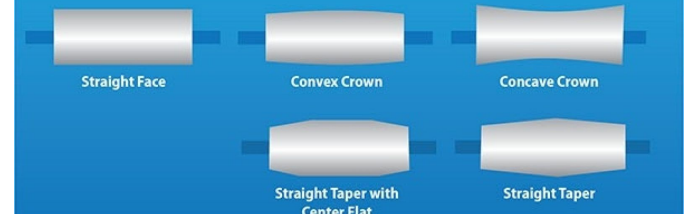
Grooved Pulley: Uses a groove to guide a rope, cable, or V-belt and prevent lateral movement.

Timing Pulley: Uses teeth that engage a timing belt to prevent slipping and maintain precise motion.

Proper belt/cable tension is necessary to prevent slipping, jumping, or excessive wear.



Crowned Pulley Options

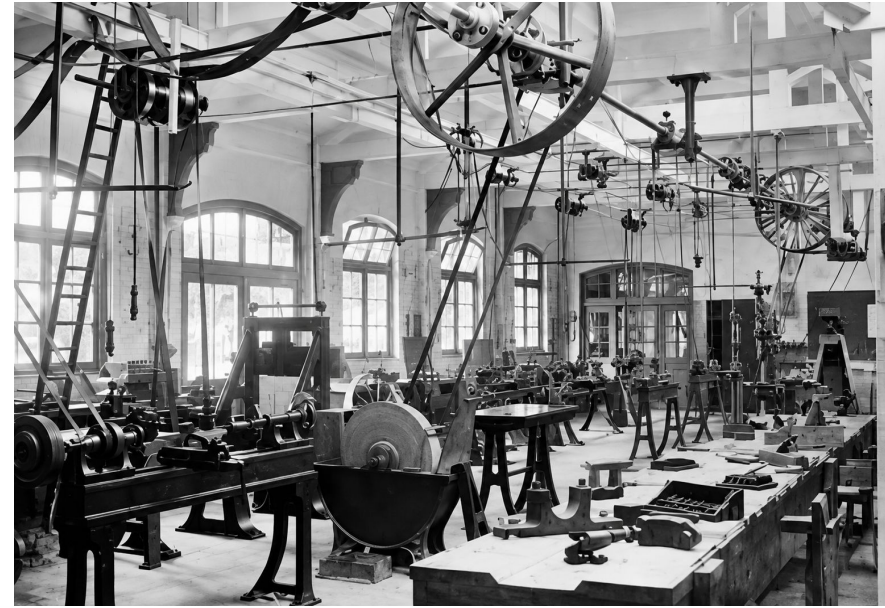
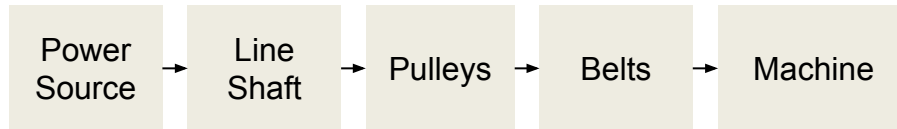


Pulleys in Industry



Line Shaft Systems - "Before individual electric motors"

- Centralized power rotated a long line shaft
- Pulleys were mounted at different locations shaft
- Flat belts transferred rotation from the shaft to individual machines.
- Pulley diameter would be varied to adjust for different machine speeds
- One power source could operate multiple machines at once!



Ratchet & Pawl



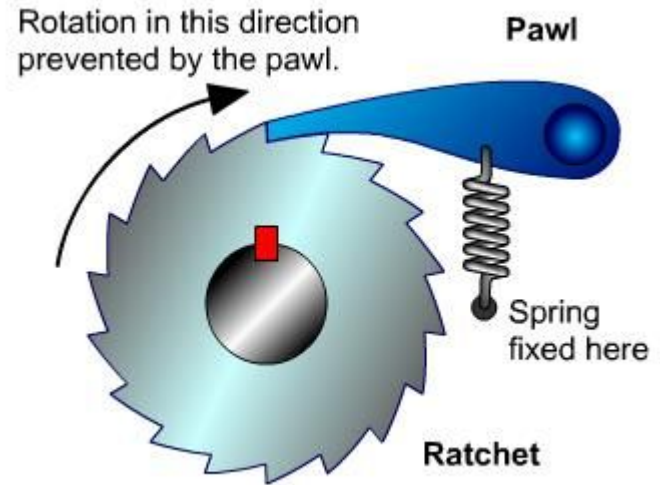
Components: Toothed ratchet wheel, a pawl, and typically a spring to maintain engagement.

Operation: The pawl slides over the angled teeth during forward rotation and catches against the steep tooth face if the ratchet attempts to rotate backward, allowing rotation in one direction while preventing rotation in the opposite direction.

Advantages: Can mechanically hold a load/position without continuously applying torque. Ratchet tooth spacing can be used to create discrete, step-by-step motion rather than continuous rotation.

Drawbacks: Tooth and pawl undergo repeated impact loads; the mechanism also produces noise and wear during engagement.

Common Applications: Winches, lifting mechanisms, tensioners, indexing systems, hand tools, and locking mechanisms.



Ratchet & pawl mechanisms turn two-way rotational freedom into controlled one-way motion.



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Springs

Springs



Springs store mechanical energy when deformed and release that energy as the spring returns toward its original position.

Hooke's Law: Spring force is proportional to displacement within the spring's elastic range:

$$\mathbf{F = kx}$$

F = spring force (N)

k = spring constant/stiffness (N/m)

x = compression or extension (m)

Stiffness: A larger **spring constant k** means more force is required to compress or extend the spring the same distance.

Preload: Springs can be intentionally compressed or stretched during assembly so they apply a force even at the mechanisms starting position.

Common Applications: Return mechanisms, counterbalancing, shock absorption, tensioning, compliant mechanisms, and maintaining contact between components.

Spring Types



Type	Typical Use
Compression	Resists being compressed
Extension	Resists being stretched
Torsion	Resists rotation
Leaf	Bending / suspension
Constant-Force	Provides nearly constant force over its travel



Compression



Extension



Torsion



Leaf



Constant-Force



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Linkages

Scissor Lift



Purpose: Cross-linked (X-shaped) folding arms provide **vertical linear motion**. Horizontal compression at the base drives vertical platform rise

Components

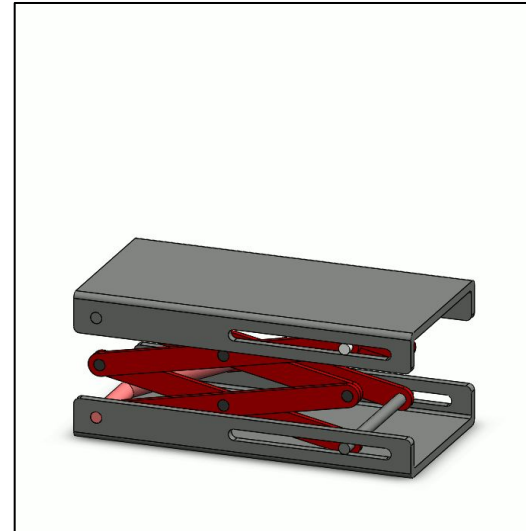
- Scissor arms, pivot pins, fixed base, moving platform, linear actuator (hydraulic or

Advantages

- Large vertical travel from compact folded height
- Stable, symmetric load distribution
- Stackable for greater height

Applications

- Aerial work platforms, car lifts
- Hospital beds, warehouse equipment



Four-Bar Linkage



Purpose: Uses connected rigid links to convert an input motion into a controlled output motion or path.

Components

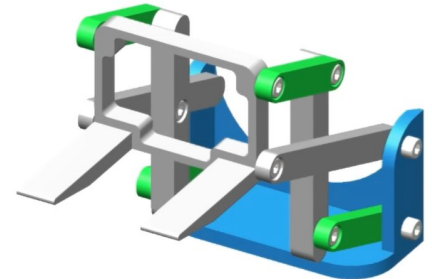
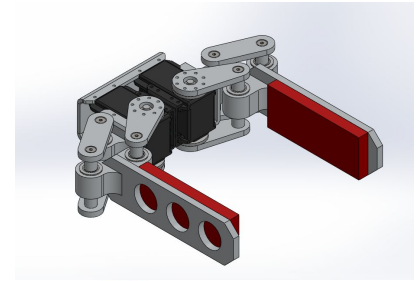
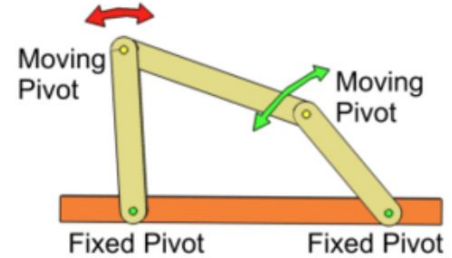
- Ground Link – fixed link connecting the two stationary pivots
- Input Link (Crank) – receives motion from the motor/actuator
- Coupler Link – connects the input and output links
- Output Link (Rocker) – produces the desired output motion
- 4 Revolute/Pivot Joints – allow rotation between the links

Advantages

- Can change motion, direction, speed, and mechanical advantage
- Motion can be customized by changing lengths and pivot locations
- Multiple components can be mechanically synchronized

Applications

- Robotic grippers, Folding/deployable mechanisms, Steering mechanisms



Slider-Crank Linkage



Purpose:

Converts rotary motion into linear reciprocating motion (vice versa)

Components

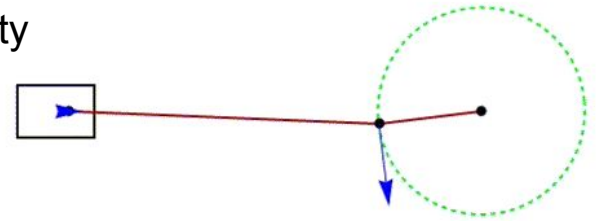
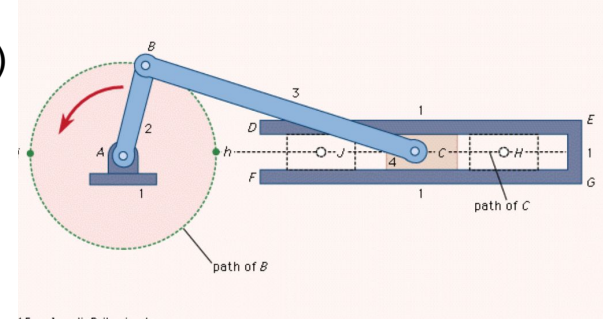
- Crank: Rotating input link
- Connecting Rod: Transfers motion
- Slider: Moves back and forth linearly
- Frame/Guide: Constrains slider motion

Advantages

- Simple & Reliable with few components and joints
- Changeable stroke and motion geometry allows for versatility

Applications

- Engines: converts piston motion crankshaft rotation
- Machinery: Produces repetitive back-and-forth motion



Parallel Motion Linkage



Purpose: Keeps output link parallel to ground link to preserve end-effector orientation throughout the full range of motion

Components

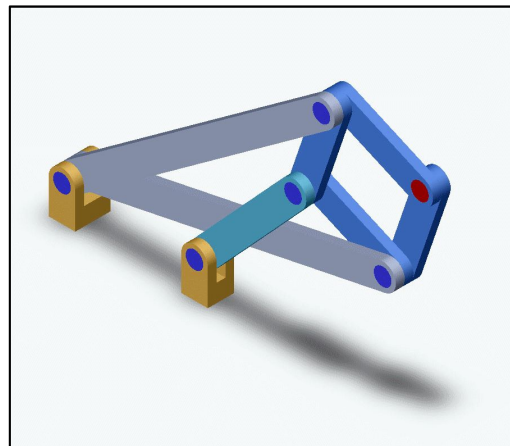
- Ground link: Fixed base of the mechanism
- Two equal-length cranks: Identical length arms connecting ground to coupler
- Coupler: Output bar (remains parallel to ground link)

Advantages

- Maintains end-effector orientation

Applications

- Robotic Arms: Keeps gripper/end-effector horizontal
- Camera Arms: Preserves camera orientation
- Architect Desk: Maintains table angle/orientation



Toggle Linkage



Purpose: Create high mechanical advantage & locking force near the toggle position

Components

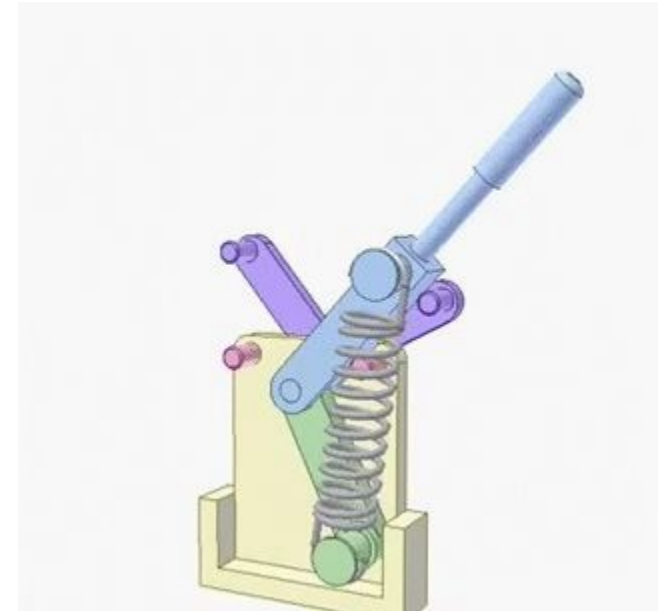
- Input link: applies driving force
- Toggle Links: pivot transmit & amplify force
- Joints/Pivots: connect moving links

Advantages

- Locking: can hold position with little input force
- Uses few mechanical components compared to others
- Produces high force in a small mechanism

Applications

- Latches: locking doors, covers, & fixtures
- Machinery like punching & forming mechanism



Compliant Mechanisms



Purpose: Uses flexible deformation instead of traditional joints to create controlled motion

Components

- Flexible Members: Bend/Deform for motion
- Rigid Sections: Transfer Forces & Guide Movement

Advantages

- Can be made in much fewer parts
- No joints required; ex: Bearings, Pins, ect..
- Doesn't require maintenance like jointed mechanisms

Applications

- Clips: Snap-fits & locking features
- Grippers: flexible robot mechanisms





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Fasteners & Joints

Fasteners



Purpose: Create re-movable mechanical joints by producing clamping force between components

Most Common Fasteners:

- Machine Screws
- Bolts
- Self-Tapping Screws
- Heat Inserts

Choosing a Fastener Depends on:

- Type
- Thread
- Diameter
- Location
- Length

Applications:

- Structural Connections
- Component Assemblies
- Anything!

Socket Head Cap Screw



- High strength
- Compact head
- Common general-purpose screw

Button Head Screw



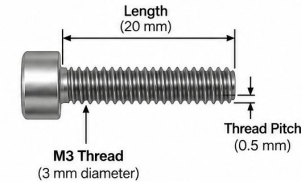
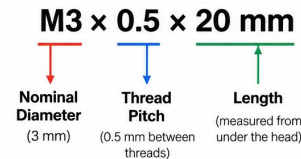
- Lower profile head
- Used when clearance is limited
- Common in lightweight assemblies

Heat-Set Insert (HI)



- Provides metal threads in plastic
- Installed by heating and pressing in
- Allows reusable threaded connections
- Common in 3D-printed parts

M3 Screw Size Example



Heat-Set Insert in Plastic



Installed in 3D-printed part
(cross-section view)

Heat Set Inserts



- Threaded inserts melted into 3D-printed parts (Use the Soldering tool with het set insert tip)
 - Let the insert cool before touching it (From experience)
- 3D-printed threads/self tapping are very easy to destroy



Important information: (M3 example)

Installed Length	3.8 mm
For Min. Material Thickness	4.57 mm
Drill Bit Size	No. 8
For Maximum Hole Diameter	0.199"

Hole cutout dimensions for CAD

Note: To make installation easy, add an 8° chamfer to $\frac{1}{4}$ "Installed Length"

Bearings



Purpose: Support rotating or moving components, while reducing friction and controlling motion.

Common Types

- Ball Bearing
- Flanged Bearing
- Thrust Bearing
- Linear Bearing

How they Work: Rolling elements (balls/rollers) separate moving surfaces & lets components move with minimal friction

Applications

- Wheels & Axles
- Motor & Gearbox Shafts
- Arms & Joints

Common Bearing Types			
Ball Bearing	Flanged Bearing	Thrust Bearing	Linear Bearing
General-purpose bearing for radial loads; used in shafts, wheels, and rollers.	Ball bearing with an integrated flange for easy mounting and axial location.	Supports axial (longitudinal) loads along the shaft axis.	Allows smooth linear motion along a shaft or rail.
			



Snap-Fit Joints



Purpose: Joins parts together without additional use of fasteners.

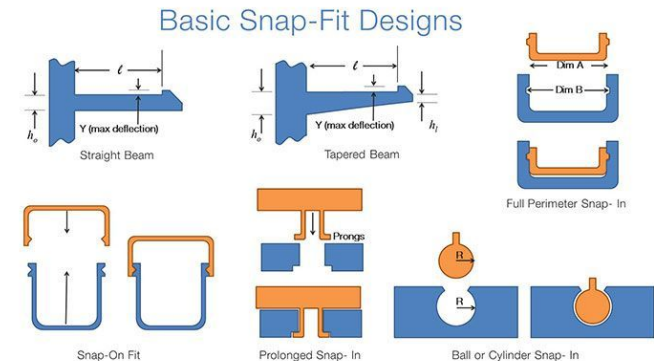
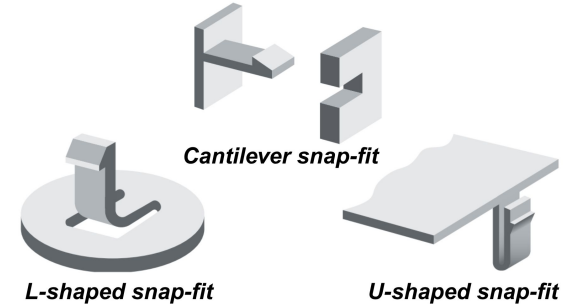
How They Work: A flexible feature temporarily bends during assembly, then returns towards its original shape and locks in position.

Common Types:

- Cantilever: Most Common; flexible arm hooks onto edge
- U-shaped: Longer flexible path allows for more deflection
- Annular: longer flexibility for shafts or cylindrical parts

Design Considerations:

- Material must allow for **ELASTIC** deformation
- Geometry: Longer/thinner features deform easier
- Stress: Avoid sharp corners that concentrate stresses
- Assembly: If the joint needs to be removable



Bambu Cut Feature



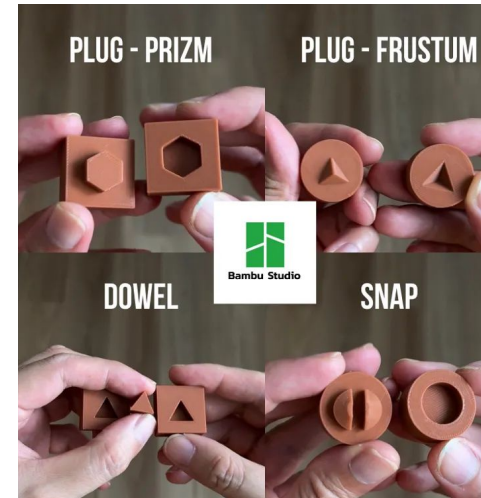
1. Tools → Cut [C] → 'Add connectors'
2. Select connector type → Confirm connectors



Tips:

- Add 0.1-0.2mm tolerance for tight fits
- Use multiple plugs/dowels to prevent rotation
- Snap fits work best for PETG or ABS – PLA may crack

Best Practice: Use plug or dovetail for structural joints, dowel when gluing, snap for removable sections



Helpful Resources



- a. [Mechanical Design Calculator \(pulleys, linear mechanisms, chain/belt\)](#)
- b. [Turtle Robotics Knowledge Base](#)
- c. [khkgears](#) - introduction to gear types
- d. [rushgears](#) - gear CAD
- e. [youtube–design for manufacturing \(DFMA\)](#)
- f. [507 movements](#)



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Next Week

Electromechanical Design



“Failure isn’t a necessary evil. In fact, it isn’t evil at all. It is a necessary consequence of doing something new.”

—Ed Catmull, Creativity Inc.

